

# A polynomial quotient bound across thin density boxes

Complete proof; independent reviews passed

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This note makes the interval widths in the positive-completion argument quantitative. In particular, no condition number of a raw truncated-power basis is used.

Let  $m \geq 2$ , and let a probability density  $\beta$  be supported on  $K$  disjoint boxes  $J_q = [c_q - \epsilon, c_q + \epsilon]$ , each of mass  $1/K$ . Assume the complementary gaps, including the two endpoint gaps, have length at least  $\Delta > 0$ . Assume also the polynomial sampling bound

$$K^{-1} \sum_q |p(c_q)|^2 \geq c_0 \|p\|_{L^2[0,1]}^2 \quad (p \in \mathcal{P}_m), \quad (1)$$

where  $c_0 > 0$  is absolute. Perturbed midpoint grids with  $K \geq Cm^4$  satisfy this assumption. No bound on  $\|\beta\|_\infty$  is needed. Set

$$S_s(a) = \int_{b>a} b^s \beta(b) db, \quad V = \mathcal{P}_m + \text{span}\{a^r S_s(a) : r + s \leq m\}.$$

Write  $G$  for the union of the complementary gaps, and  $\Pi_j$  for orthogonal projection in  $L^2(G)$  onto polynomials of degree at most  $j$  separately on each gap. Define

$$\mathcal{S}_0 = \text{span}(1, S_0, \dots, S_m), \quad \mathcal{S}_1 = \text{span}(1, a, S_0, \dots, S_m, aS_0, \dots, aS_{m-1}).$$

**Theorem 1.** *For sufficiently small absolute  $c > 0$ , if  $\epsilon \leq c\Delta/(m+1)^3$ , then for every  $F \in V$*

$$\text{dist}_{L^2[0,1]}(F, \mathcal{S}_0) \leq C(m+1)^4 \Delta^{-3/2} \|(I - \Pi_0)F\|_{L^2(G)}, \quad (2)$$

$$\text{dist}_{L^2[0,1]}(F, \mathcal{S}_1) \leq C(m+1)^6 \Delta^{-5/2} \|(I - \Pi_1)F\|_{L^2(G)}. \quad (3)$$

*In particular the kernel of restriction followed by  $I - \Pi_j$  is exactly  $\mathcal{S}_j$ , for  $j = 0, 1$ .*

*Proof.* Choose any representation  $F = P + \sum c_{rs} a^r S_s$ . Define  $T$  on this representation by differentiating  $P$  and the explicit powers  $a^r$  while leaving every  $S_s$  unchanged. For  $0 \leq j \leq m$  put

$$D_j(a) = \sum_{r \geq j, s} c_{rs}(r)_j a^{r-j+s}.$$

It has degree at most  $m - j$ , and the absolutely continuous identities

$$(T^j F)' = T^{j+1} F - \beta D_j \quad (4)$$

hold, with  $T^{m+1} F = 0$ . On the gaps,  $T^j F = F^{(j)}$ . Let

$$M_j = \|T^j F\|_{L^\infty[0,1]}, \quad E_j = \|F^{(j)}\|_{L^\infty(G)}.$$

Integrating (4) across a box gives

$$\left| \int_{J_q} \beta D_j \right| \leq 2E_j + 2\epsilon M_{j+1}. \quad (5)$$

For any  $p \in \mathcal{P}_m$ , Markov and the  $L^2$  Nikolskii inequality give  $\|p'\|_\infty \leq C(m+1)^3 \|p\|_2$ . Consequently

$$\left| K \int_{J_q} \beta p - p(c_q) \right| \leq C\epsilon(m+1)^3 \|p\|_2.$$

Assumption (1) and sufficiently small  $\epsilon(m+1)^3$  imply

$$\|p\|_2 \leq CK \max_q \left| \int_{J_q} \beta p \right|.$$

Applied to (5), followed by Nikolskii, this yields

$$\|D_j\|_\infty \leq C(m+1)K(E_j + \epsilon M_{j+1}).$$

Integrating (4) from a box endpoint to an arbitrary point of that box, and using its total mass  $1/K$ , now gives

$$M_j \leq C(m+1)(E_j + \epsilon M_{j+1}). \quad (6)$$

On each gap the usual sup-norm Markov inequality gives  $E_{j+1} \leq C(m+1)^2 \Delta^{-1} E_j$ . Starting from  $M_{m+1} = 0$ , substitute this inequality in the backward iteration of (6). Our bound on  $\epsilon$  makes the resulting geometric series summable, so

$$M_j \leq C(m+1)E_j \quad (0 \leq j \leq m). \quad (7)$$

For (2), let  $U_S$  denote the same linear combination of the  $S_s$  as the monomials in  $D_0$ . Then  $U'_S = -\beta D_0$ , so  $(F - U_S)' = TF$ . Add a constant to  $U_S$  to remove the mean of the difference. Poincare and (7) give distance at most  $C(m+1)E_1$ . Gapwise Nikolskii and inverse Markov give

$$E_1 \leq C(m+1)\Delta^{-1/2} \|F'\|_{L^2(G)} \leq C(m+1)^3 \Delta^{-3/2} \|(I - \Pi_0)F\|_{L^2(G)}.$$

This proves (2).

For (3), put  $U = D_0 - aD_1$  and  $W = D_1$ , and form  $H = U_S + aW_S \in \mathcal{S}_1$  by replacing powers by the corresponding  $S_s$ . Then

$$H' = W_S - \beta D_0, \quad H'' = -\beta D_1 - (\beta D_0)', \quad F'' = T^2 F - \beta D_1 - (\beta D_0)'.$$

These identities hold distributionally even if  $\beta$  jumps at box endpoints. Thus  $(F - H)'' = T^2 F$ . Add a global affine polynomial to  $H$  and use the second-order Poincare inequality to obtain distance at most  $CM_2 \leq C(m+1)E_2$ . Finally,

$$E_2 \leq C(m+1)\Delta^{-1/2} \|F''\|_{L^2(G)} \leq C(m+1)^5 \Delta^{-5/2} \|(I - \Pi_1)F\|_{L^2(G)}.$$

Here subtracting the affine projection on each gap leaves  $F''$  unchanged. This proves (3). The reverse kernel inclusions are immediate because every  $S_s$  is constant on every gap.  $\square$

**Corollary 1** (Mass-and-first-moment positive completion). *Assume  $\beta$  is rational piecewise polynomial and all endpoints are rational. Suppose  $g$ , extended by zero on the boxes, is a rational piecewise affine function on the gaps, satisfies  $1/2 \leq g \leq 3/2$  there, and*

$$\int_0^1 (1 - g)H = 0 \quad (H \in \mathcal{S}_1).$$

*If*

$$C(m+1)^7 \Delta^{-3} \|1 - g\|_{L^2[0,1]} \leq 1/4,$$

*then there is a rational piecewise polynomial density  $f$ , supported on the gaps, with  $1/4 \leq f \leq 7/4$  on  $G$ , having the same mass and first moment as  $g$  on every gap, and satisfying*

$$\int fF = \int F \quad (F \in V).$$

*Every polynomial piece has degree at most  $m$ .*

*Proof.* On  $W = (I - \Pi_1)V \subset L^2(G)$  define  $\Lambda((I - \Pi_1)F) = \int(1 - g)F$ . The kernel statement and assumed compatibility make this well-defined. Theorem 1 gives its norm at most  $C(m + 1)^6 \Delta^{-5/2} \|1 - g\|_2$ . Its finite-dimensional Riesz representative  $h \in W$  has this same norm and has zero mass and first moment on each gap. Gapwise Nikolskii gives  $\|h\|_\infty \leq C(m + 1) \Delta^{-1/2} \|h\|_2 \leq 1/4$ . Set  $f = g + h$ . All assertions follow. A rational basis for  $W$  has rational Gram matrix and right side, so the unique Riesz representative has rational coefficients.  $\square$

**Theorem 2** (A fixed number of cell moments). *Fix  $q \geq 0$  and suppose  $m \geq \max(2, q)$ . Retain all box, sampling, and width assumptions above. Define*

$$\mathcal{S}_q = \mathcal{P}_q + \text{span}\{a^r S_s(a) : 0 \leq r \leq q, r + s \leq m\},$$

*and let  $\Pi_q$  project onto  $\mathcal{P}_q$  separately on each gap. Then*

$$\text{dist}_{L^2[0,1]}(F, \mathcal{S}_q) \leq C_q(m + 1)^{2q+4} \Delta^{-q-3/2} \|(I - \Pi_q)F\|_{L^2(G)} \quad (F \in V).$$

*In particular the kernel is exactly  $\mathcal{S}_q$ . For the completion statement assume additionally that  $\beta$  is rational piecewise polynomial and all endpoints are rational. If a rational piecewise polynomial  $g$  of degree at most  $q$  on the gaps, extended by zero on the boxes, has values between  $1/2$  and  $3/2$  and satisfies  $\int(1 - g)H = 0$  for all  $H \in \mathcal{S}_q$ , then the condition*

$$C_q(m + 1)^{2q+5} \Delta^{-q-2} \|1 - g\|_{L^2[0,1]} \leq 1/4$$

*suffices for a rational positive completion  $f$ , with the same first  $q$  moments as  $g$  on every gap and with  $\int fF = \int F$  for all  $F \in V$ . On each gap  $f$  is polynomial of degree at most  $m$  and lies between  $1/4$  and  $7/4$ .*

*Proof.* The already proved backward recurrence and bound  $M_j \leq C(m + 1)E_j$  hold in all derivative orders through  $m$ . For a generator  $a^r S_s(a)$ , subtract

$$H_{r,s}(a) = \int_{b>a} b^s \left[ \sum_{l=0}^{\min(q,r)} \binom{r}{l} b^{r-l} (a - b)^l \right] \beta(b) db.$$

Expanding the bracket shows that it belongs to  $\mathcal{S}_q$ : each term is a constant times  $a^j S_{r+s-j}(a)$  with  $j \leq q$ . The bracket is the degree- $q$  Taylor polynomial of  $a^r$  about  $a = b$ . The residual kernel therefore has zero boundary derivatives through order  $q$ . Repeated differentiation under this Volterra integral, or its distributional version for merely integrable  $\beta$ , gives

$$(a^r S_s - H_{r,s})^{(q+1)} = T^{q+1}(a^r S_s).$$

Apply the construction to the chosen representation of  $F$  and leave its ordinary polynomial part in the remainder. We obtain  $H \in \mathcal{S}_q$  with  $(F - H)^{(q+1)} = T^{q+1}F$ . Subtract a global degree- $q$  polynomial to remove the first  $q + 1$  initial Taylor data of  $F - H$ . The order- $(q + 1)$  Poincare inequality then bounds its  $L^2$  norm by  $C_q M_{q+1}$ . For  $q = m$  the remainder is identically zero; otherwise the recurrence bounds this by  $C_q(m + 1)E_{q+1}$ . Gapwise polynomial evaluation and inverse Markov give

$$E_{q+1} \leq C(m + 1) \Delta^{-1/2} \|F^{(q+1)}\|_{L^2(G)} \leq C_q(m + 1)^{2q+3} \Delta^{-q-3/2} \|(I - \Pi_q)F\|_{L^2(G)}.$$

This proves the quotient and its kernel assertion.

For completion repeat the Riesz argument on  $W = (I - \Pi_q)V$ . The functional induced by  $\int(1 - g)F$  is well-defined by the kernel assertion and compatibility. Its representative has norm at most the displayed quotient constant times  $\|1 - g\|_2$  and zero moments through degree  $q$  in each gap. Gapwise evaluation costs only  $C(m + 1) \Delta^{-1/2}$ , giving the stated supremum bound. Add this representative to  $g$ . Rationality follows from a rational basis, Gram matrix, and right side, exactly as in Corollary 1.  $\square$

**How compatibility is checked.** If  $g$  has cumulative mass  $r_{0q}$  and cumulative first moment  $r_{1q}$  just before box  $q$ , then compatibility on the generators  $S_s$  and  $aS_s$  is respectively

$$\sum_q r_{0q} \int_{J_q} b^s \beta(b) db = \int b^{s+1} \beta(b) db, \quad \sum_q r_{1q} \int_{J_q} b^s \beta(b) db = \frac{1}{2} \int b^{s+2} \beta(b) db.$$

The remaining two conditions are total mass 1 and first moment  $1/2$ . All of these are linear weighted-moment identities.

**Polynomial scales suffice.** On a gap of length at least  $\Delta$ , the affine density with prescribed mass and first moment differs from 1 by at most  $C\Delta^{-2}$  times the discrepancies of these two moments. Thus if the cumulative data are within  $\eta$  of uniform prefix data and the box centers within  $\zeta$  of the reference nodes, then

$$\|g - 1\|_{L^\infty(G)} \leq C\Delta^{-2}(\eta + \zeta + \epsilon), \quad \|1 - g\|_{L^2[0,1]} \leq C\Delta^{-2}(\eta + \zeta + \epsilon) + \sqrt{2K}\epsilon.$$

When  $K, \Delta^{-1}$  are polynomial in  $m$ , choosing successively small fixed inverse powers for  $\eta, \epsilon, \zeta$  meets all hypotheses. This yields polynomially small boxes, rather than an unspecified continuity scale.